#DPL2026

# DATA PREMIER LEAGUE 2026

*Decode the Game. Build the Analytics.*

## PRELIMINARY ROUND — SUBMISSION

**TEAM / PARTICIPANT NAME**

Faizan Khan

**QUESTION ATTEMPTED**

Question 1 — Franchise Report

**SELECTED IPL TEAM(S) / SEASONS**

Royal Challengers Bengaluru · IPL 2023 and 2024 · chosen by a screen of all ten teams, then audited decision by decision

**SUBMISSION DATE**

16 / 10 / 2026

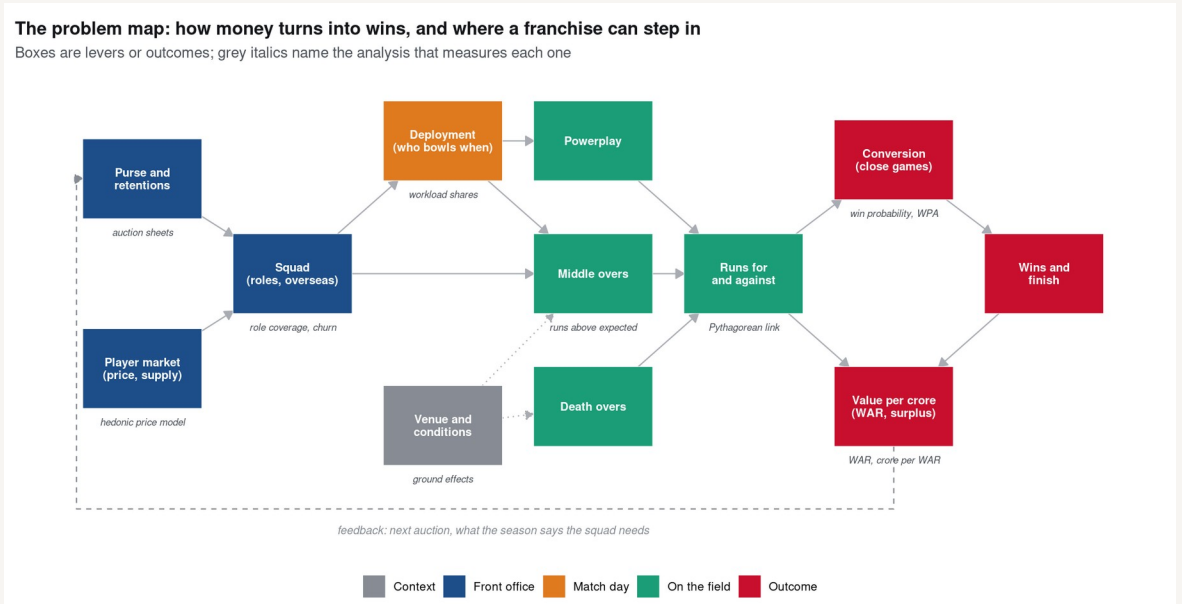
SECTION 1

REQUIRED

# Context & Analytical Objective

### Two problems inside one question

- **Problem A, which team?** Every team went through three gates fixed in advance: under-converted in both seasons, a weakness that survives three hurdles, and a money problem. Only RCB passes all three.
- **Problem B, that team.** What is wrong, what the franchise did about it at the auction, how that turned out, and what to do next.
- **Key question.** Which part of RCB's cricket was persistently below par, did the December 2023 auction fix it, and what is the cheapest fix now?
- **Approach.** Situation, search, obvious gap, representation, hypotheses, key question, run once for each problem.




*The problem map: money buys a squad, the squad is deployed into phases, phases produce runs, runs become wins, and value per crore feeds the next auction. Every analysis measures one link.*

SECTION 2

REQUIRED

# Data & Methodology

## Data Sources

- DPL ball-by-ball, 2023–24: 34,967 rows × 61 fields, 145 matches, 14 venues
- Ball tracking: line, length, shot, control, wagon wheel
- Auction sheets for both seasons: retained, sold, traded, unsold (399 priced player-seasons)
- Mentor sessions and published work: cricWAR, T20 Basics, Runs Above Expected, Malhotra

## Cleaning & Prep

- bat_out is TRUE on 99.9% of rows: dismissals rebuilt; 1,797 true wickets
- Retired hurt is not a wicket; wagon zones decoded (99.2% agreement)
- All game state rebuilt as it stood before each ball; a guard blocks post-ball features
- Every comparison made within season (2024 scoring jumped under the impact-player rule)

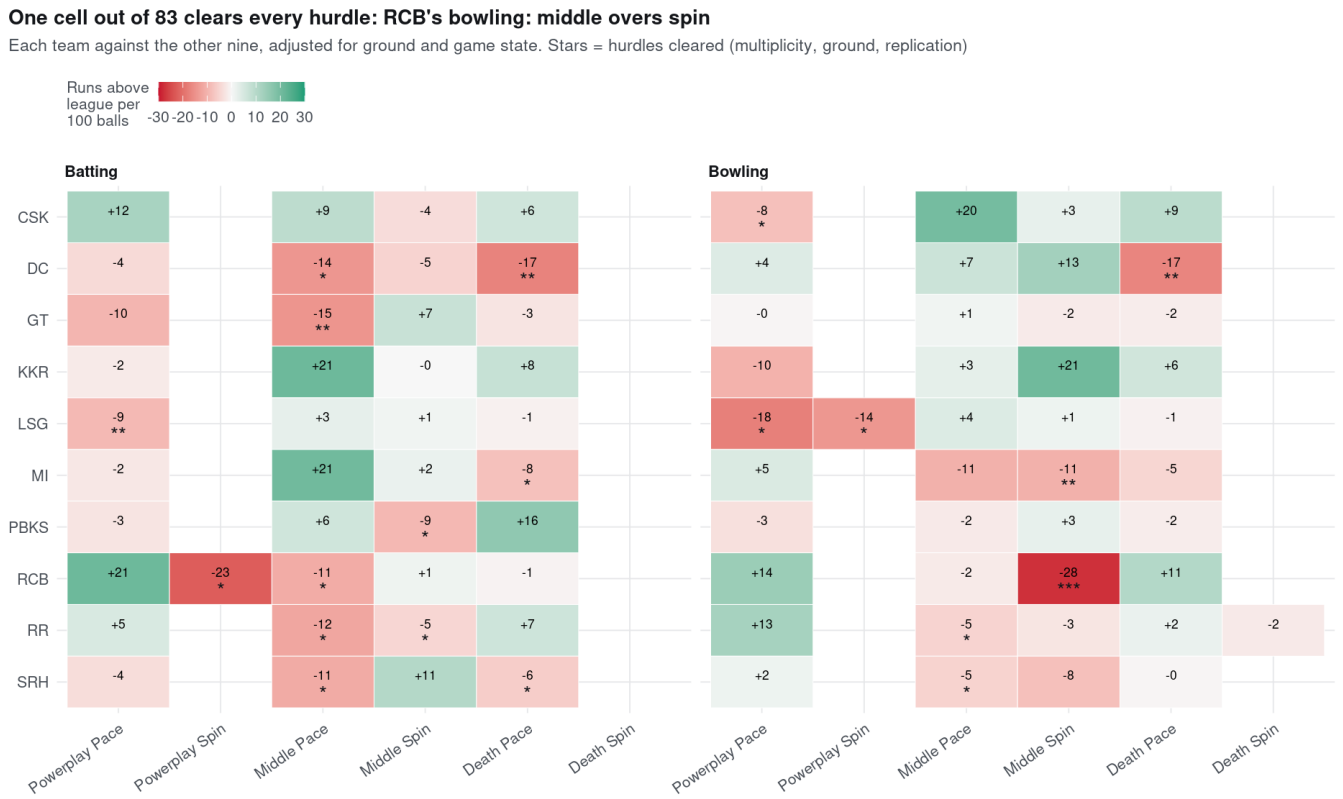
## Analytical Methods

- League EDA: 22 views by over, phase, ground, line/length, shot, matchup, team and price
- Problem A: Pythagorean gate, 12-cell scan with BH, ground fixed effects and replication, money gate
- Tools: expected runs M0→M4, win probability W0→W4, WAR, crore per WAR, hedonic prices, pressure
- Problem B: eight written hypotheses; decision audit 2023 → auction → 2024, foresight vs hindsight

Code: github.com/sportsanalystf/data-premier-league-26/tree/main/01-team-diagnostics · Rscript run_all.R rebuilds every number (seed 2026) · Rscript build.R rebuilds the reports, paper, dashboard and deck

SECTION 3

# Performance Overview




*Every team, every cell: runs above the rest of the league per 100 balls, ground and game state held fixed. Stars count hurdles cleared.*

**1 of 10**

Teams passing all three gates: RCB only

**−27.9**

Middle-overs spin, runs per 100 balls vs the rest: −26.8 in 2023, −29.0 in 2024

**₹36 : ₹2 cr**

New auction money for 2024 on pace against spin

SECTION 4

REQUIRED

# Detailed Analytical Findings

### 1 RCB was chosen by the data, on three separate gates

Below Pythagorean expectation in both seasons (−5.4 and −10.9 points); the only one of 83 team-cells to clear BH (q = 0.061), ground fixed effects (p = 0.0014) and replication; and the league's most negative surplus (−₹11.7 cr). Punjab passed two gates; nobody else more than one.

### 2 The leak is middle-overs spin, not bowling in general

With the ground held fixed the bowling gap falls from +5.3 to −0.6 per 100 balls, and powerplay and death bowling both rank first in the league. Middle-overs spin is −27.9 against the rest, costing 105 runs (1.2 wins) in 2023 and 103 in 2024. No frontline spinner was above the league median.

### 3 The auction went the other way, and 2024 turned elsewhere

Both main spinners were released; ₹36.2 cr of new money went on pace, ₹2.0 cr on spin. The 2024 turn (1/7 to 6/7) came from the powerplay (wickets 4 → 18); middle-overs spin stayed the league's worst. Clutch does not repeat (r = −0.28).

SECTION 5

# Strengths & Weaknesses of Chosen Methodology

## STRENGTHS

- **Fair choice of team.** Three gates, fixed in advance, every team treated the same; nothing added up into one score.
- **Like-for-like.** Ground, over, wickets, bowler type and set-ness held fixed; ground also tested with fixed effects.
- **Honest validation.** Train 2023, test 2024. The simplest well-calibrated models won (M2 slope 1.004; W1 log-loss 0.508).
- **Decisions in order.** The audit separates what was knowable at the time (foresight) from what we know now (hindsight).
- **Checked against results.** Squad WAR tracks wins (r = 0.70) without seeing a result.

## LIMITATIONS

- **Two seasons.** Gate 1 holds in only 58% of match resamples; BH q for the cell is 0.061, so at a 5% level no team clears gate 2.
- **Heavy shrinkage.** k ≈ 585 balls for bowlers: projected gains (+0.24 wins) are far below the measured cost.
- **Thin foresight market.** Only 6 spinners in the Dec 2023 pool had 2023 middle-overs evidence.
- **Not causal.** A released player doing well elsewhere mixes skill with a new team, role and ground.
- **Incomplete prices.** Some replacement signings are not on the auction sheets.

SECTION 6

REQUIRED

# Interpretation & Practical Significance

## Cricket interpretation

- At Chinnaswamy raw economy punishes every bowler; adjusted, the new-ball and death attack were assets.
- Overs 7–15 decide whether a side reaches the death with wickets in hand. Middle-overs wickets go with winning more than anything else (r = 0.69).
- RCB's spinners in that phase were the league's worst two seasons running; Karn Sharma still bowled 39% of the overs in 2024.
- Close games went 0–3: variance, not character. Clutch does not repeat.

## Practical relevance to the franchise

- Overseas players cost about +170% for the same role; Indian bowlers are the cheapest wins.
- The proven middle-overs spinners were all retained elsewhere: the route is a trade, the route used for Cameron Green (₹17.5 cr).
- In hindsight an established spinner in RCB's 2024 overs was worth about a win; Axar Patel 1.5 (₹9.0 cr).
- Build around Yash Dayal, Mohammed Siraj, Dinesh Karthik and Rajat Patidar; review Glenn Maxwell and Alzarri Joseph on price.

SECTION 7

# Conclusion & Key Takeaways

### Buy one Indian middle-overs spinner, on more than one season

Top-quarter shrunk rate (≥ +5.7 per 100); bid ceiling = projected WAR × ~₹6.0 cr. Archetype: Varun Chakravarthy, Axar Patel, Krunal Pandya, Harpreet Brar. Trade if the pool is thin.

### Move the middle overs now, for free

More for Cameron Green, Mohammed Siraj and Glenn Maxwell, fewer for the bottom of the list; track middle-overs spin every five matches.

### Don't buy ‘bowling’ or ‘clutch’

The new ball and the death are already strong once the ground is removed, and clutch does not repeat. Keep the big bids and overseas slots for proven overall rate.

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# References & Code Repository

## References


Bandyopadhyay, Runs Above Expected, arXiv:2608.18020 · Ganjoo, T20 Basics · Rafique, cricWAR (CMSAC 2022)
Malhotra, J. Sports Analytics 8(3) 2022 · Date, A Cricketing View · Nielsen, Cricket Sims · Weston, IPL Auctions
Benjamini & Hochberg (1995) · DPL 2026 mentor sessions: A. Desai, O. Walunj, S. Gupta, H. Ganjoo
Epilogue (outside the data): Sportskeeda and Onmanorama on the 2025 auction; cricket.com on the 2025 final


## Code, Reproducibility & Video Explanation Links

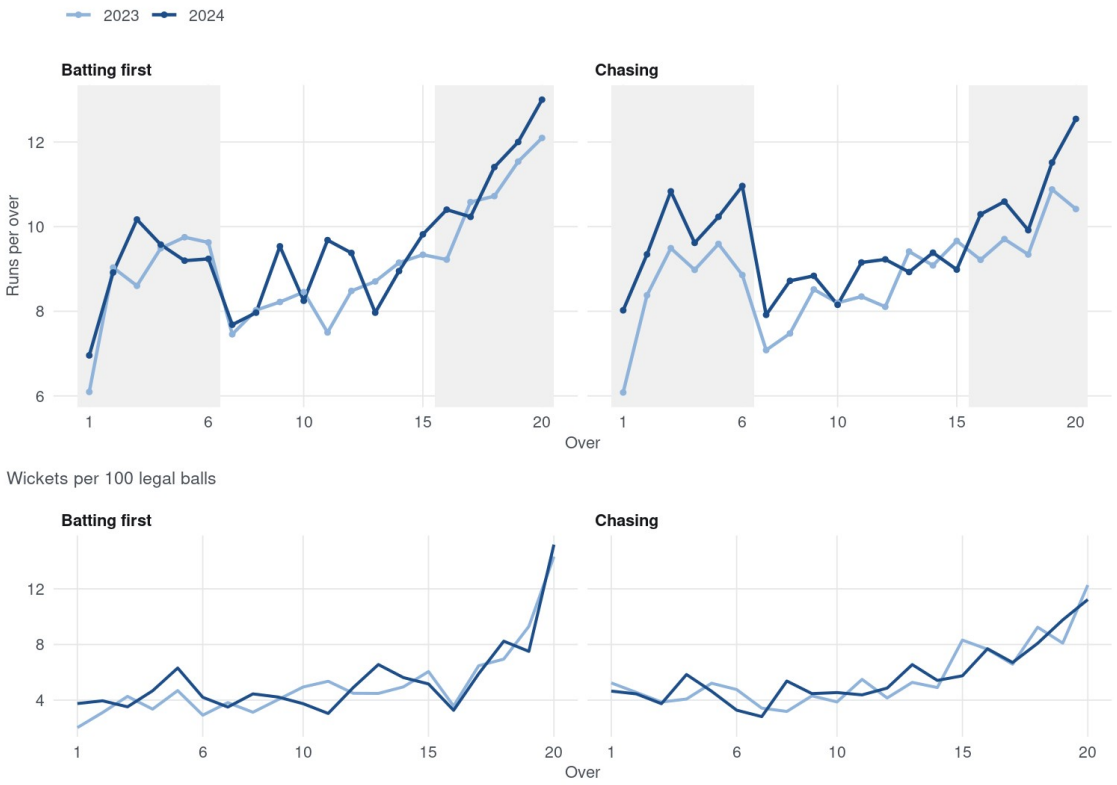
**Repository:** github.com/sportsanalystf/data-premier-league-26/tree/main/01-team-diagnostics
**Rebuild:** Rscript build.R (run_all.R → figures → numbers → three reports → paper → dashboard → deck); R 4.3, Quarto; seed 2026
**Deliverables:** report/01–03 (EDA, team review, player appendix), paper/paper.pdf, appendix/dashboard.html, data/, this deck
**Video explanation:** video/DPL2026_Q1_FaizanKhan_video.mp4 in the repository (7 min, subtitled)

APPENDIX — NOT COUNTED IN SLIDE LIMIT

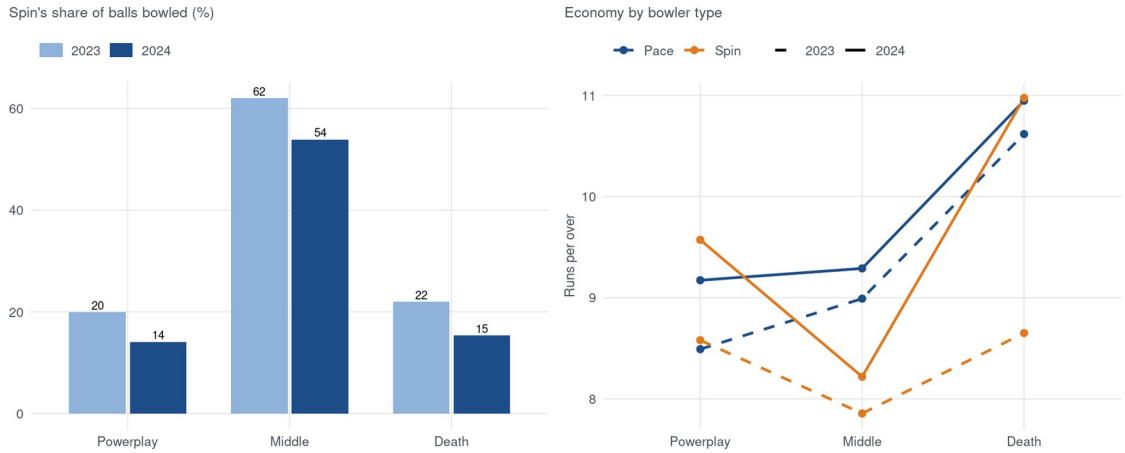
# A1 · The league: how an innings is shaped

**Most overs got dearer in 2024, the chase powerplay most of all; the shape of an innings did not change**

Runs per over, league-wide (shaded: powerplay and death)




**Teams bowled less spin in 2024, yet in the middle overs it was still a run an over cheaper than pace**



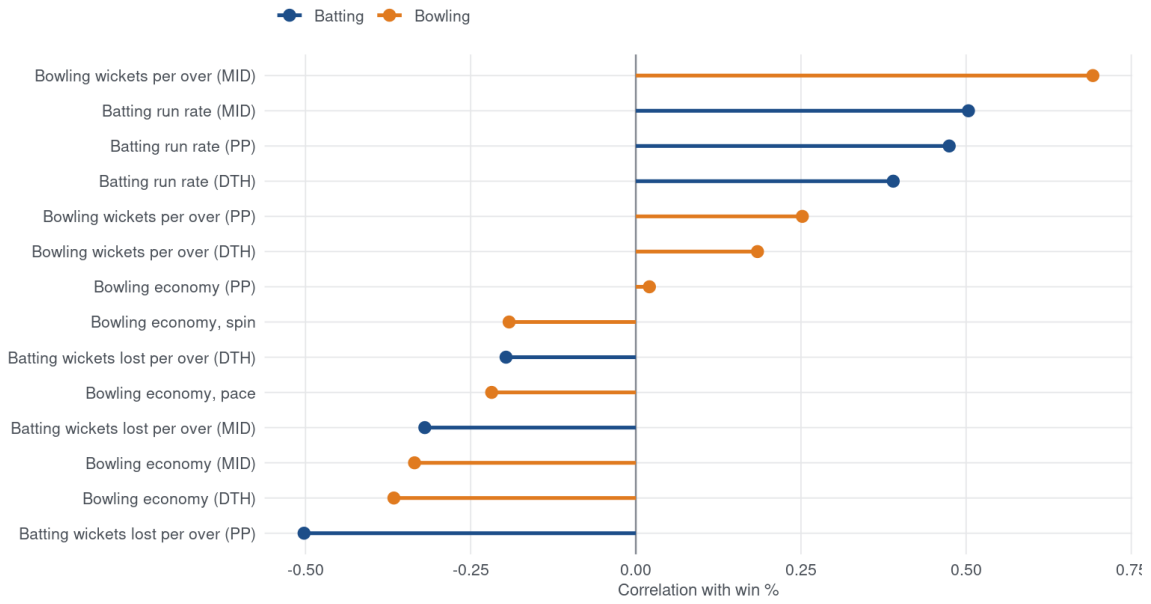

*Scoring rose from 8.94 to 9.52 an over, mostly in the powerplay and at the death (middle overs +0.41). Spin is the middle-overs tool and stayed cheaper than pace there in 2024.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A2 · The league: what goes with winning, and where

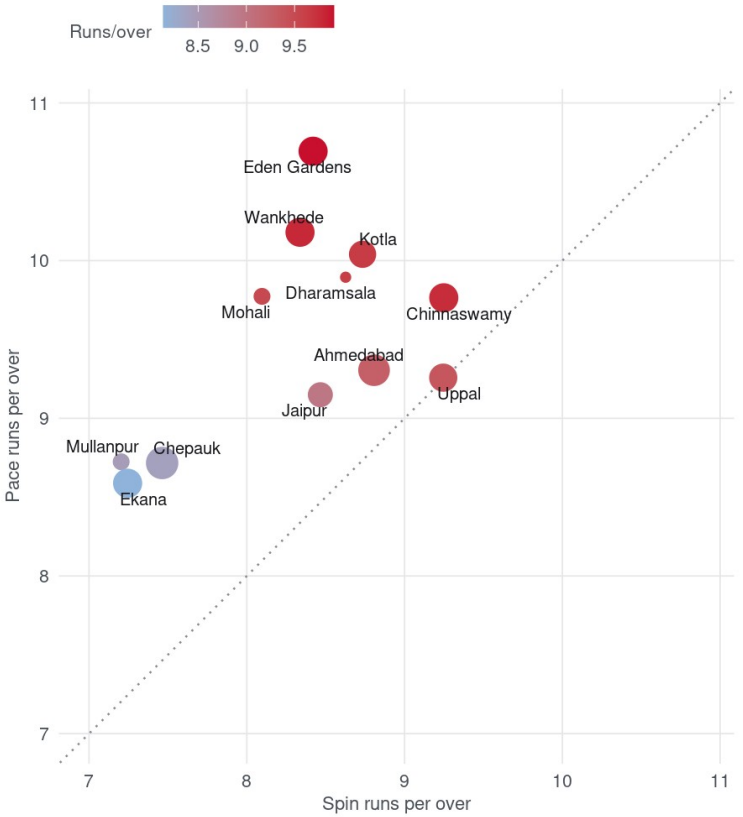
**Taking middle-overs wickets is the team habit most closely tied to winning**

Correlation with win %, 20 team-seasons, each rate centred within its season




**Grounds differ by 1.8 runs an over, and spin is cheaper than pace at every one**

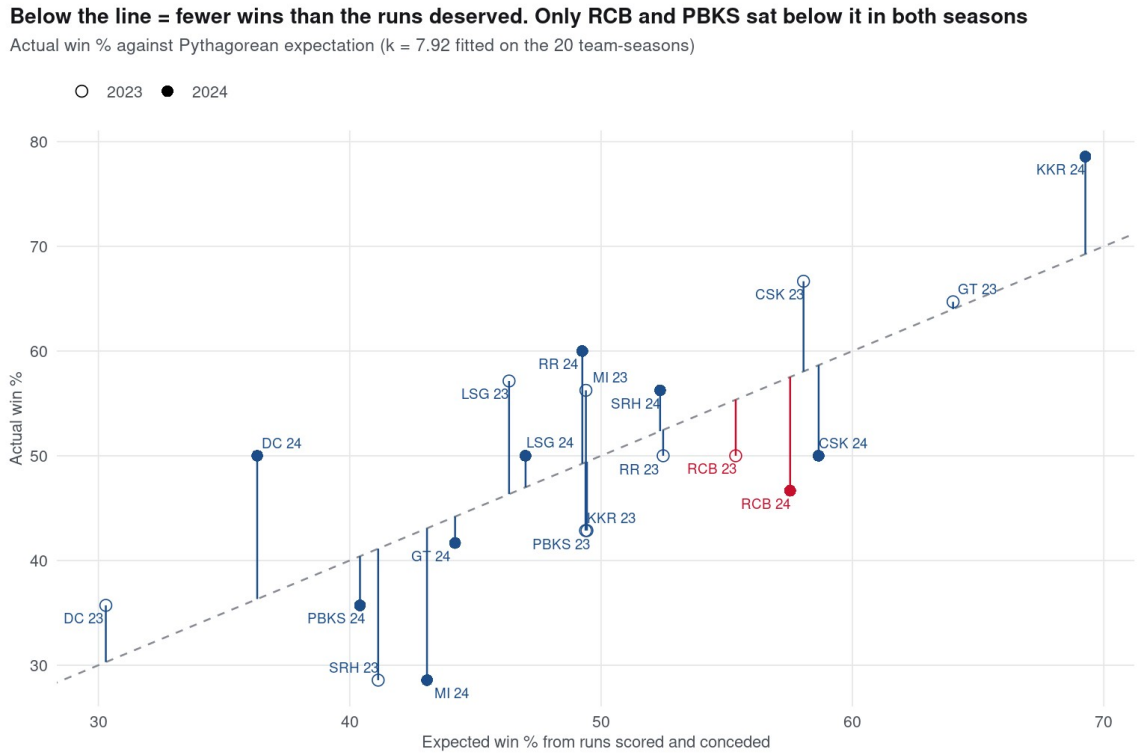
Economy of pace (y) against spin (x) at each ground with 4+ matches; dotted line: equal

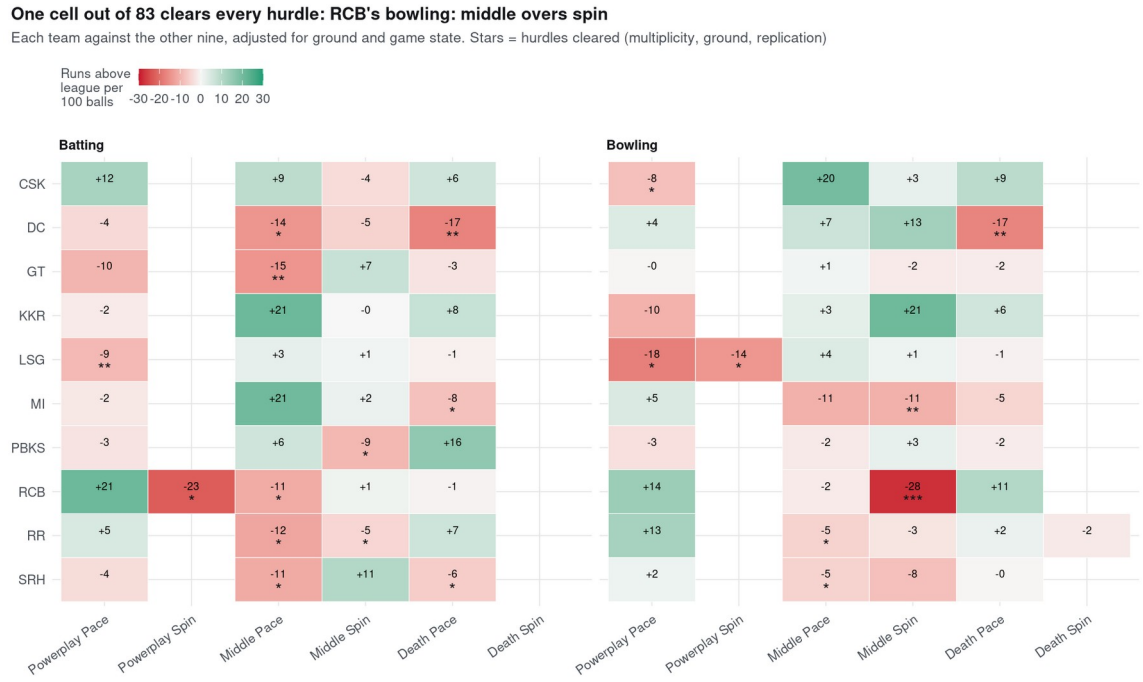



*Middle-overs wickets are the team habit most tied to winning (r = 0.69, 20 team-seasons). Grounds differ by 1.8 runs an over, so every comparison holds venue fixed.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A3 · Problem A: results and the cell scan



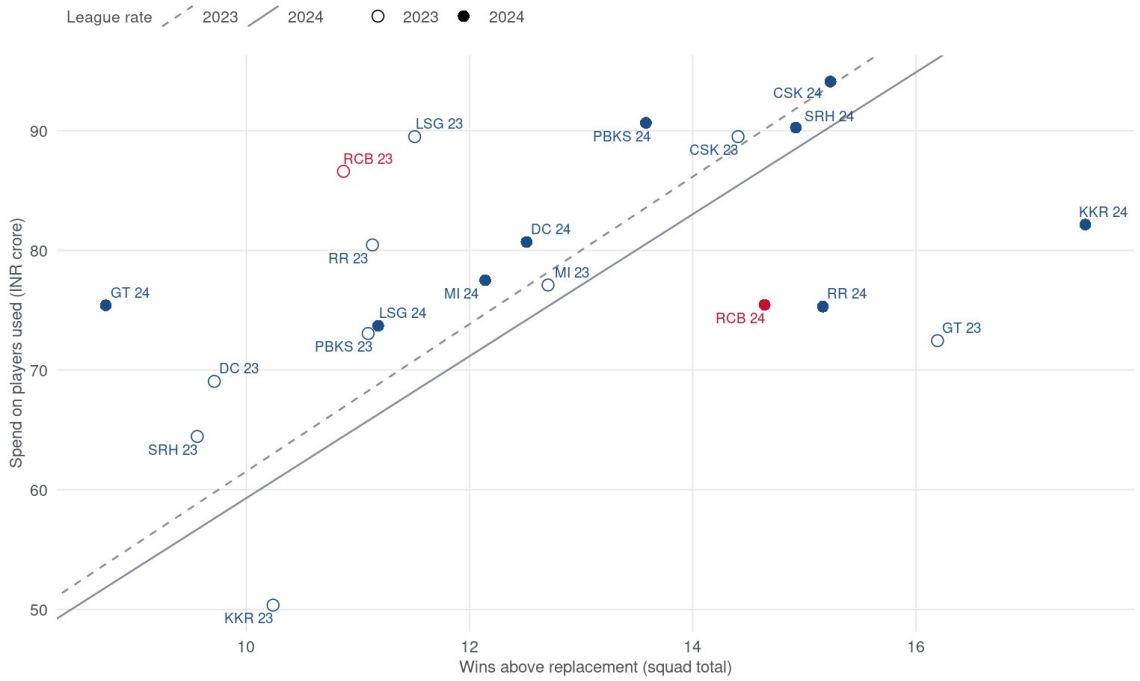

*Left: only RCB and PBKS won fewer games than their runs deserved in both seasons. Right: one cell of 83 clears all three hurdles, RCB's middle-overs spin.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A4 · Problem A: money, and every dimension

**Above the line = paid more than the wins were worth. RCB 2023 was the second-worst deal of the 20**

Squad spend on players who played against wins above replacement; lines = league crore per WAR




**Every team on every dimension: nobody is good at everything, and RCB's weak spots are specific**

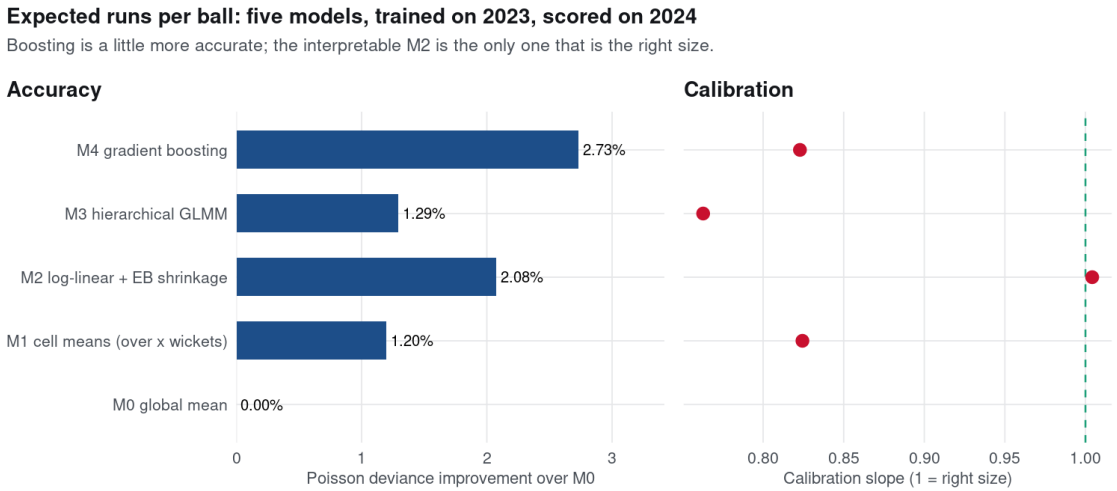
Cell values printed; colour = rank among the ten (green best). Both seasons pooled.

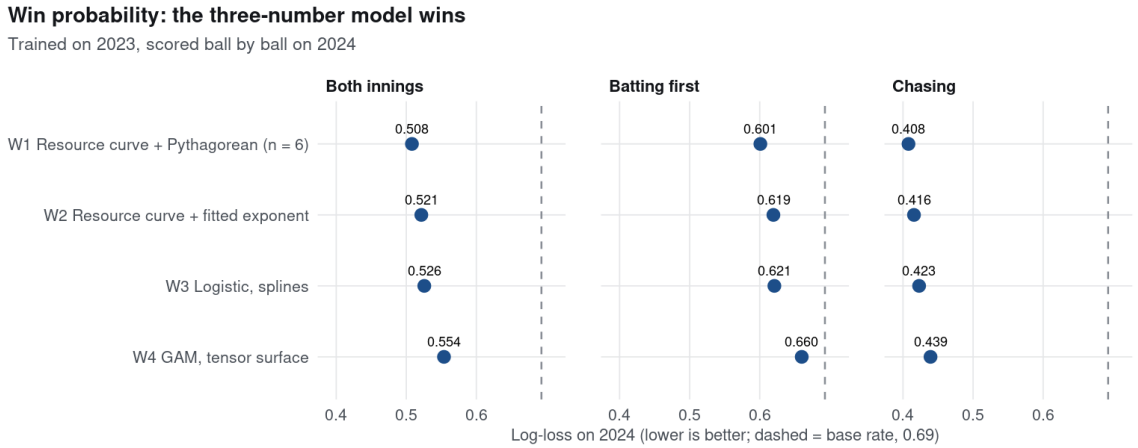



*RCB had the most negative surplus (−₹11.7 cr) over two seasons. The scorecard: powerplay and death bowling both rank first; middle-overs bowling 10; spin 10.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A5 · The measuring tools: two model ladders




*Trained on 2023, scored on 2024. Boosting is most accurate (+2.7%) but only M2 is well calibrated (slope 1.004). The three-parameter win model W1 beats logistic and GAM (log-loss 0.508).*

# A6 · From runs to wins to crore

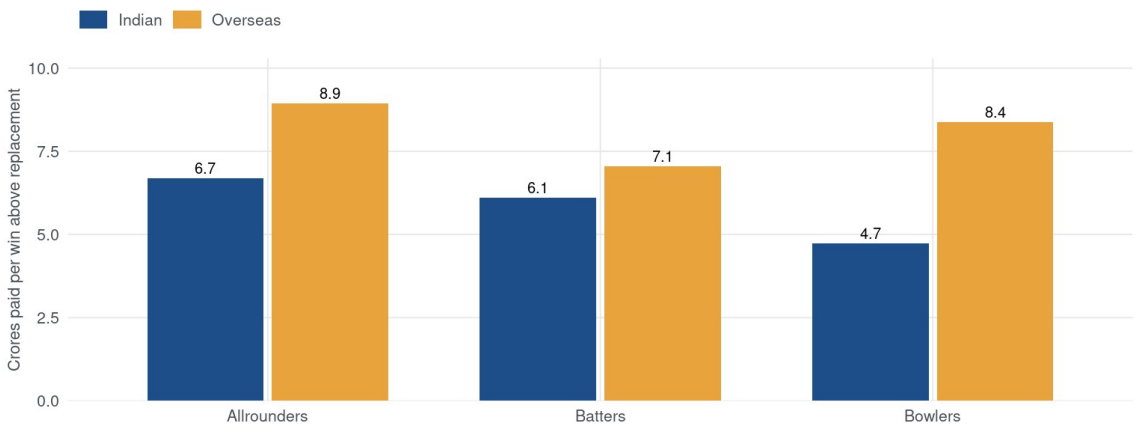
**Does the player value add up to the table?**

Team-seasons, r = 0.75. WAR is built ball by ball and never sees a result.




**The cheapest wins in the market are Indian bowlers**

Both seasons, priced players only



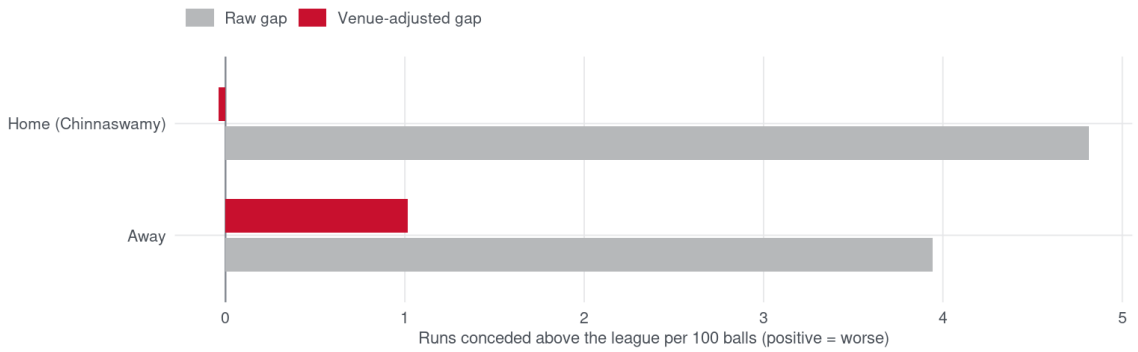

*Squad WAR, built ball by ball, tracks wins (r = 0.75); 90 runs per win. Indian bowlers are the cheapest wins in the market.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A7 · Problem B: the ground, and the middle overs

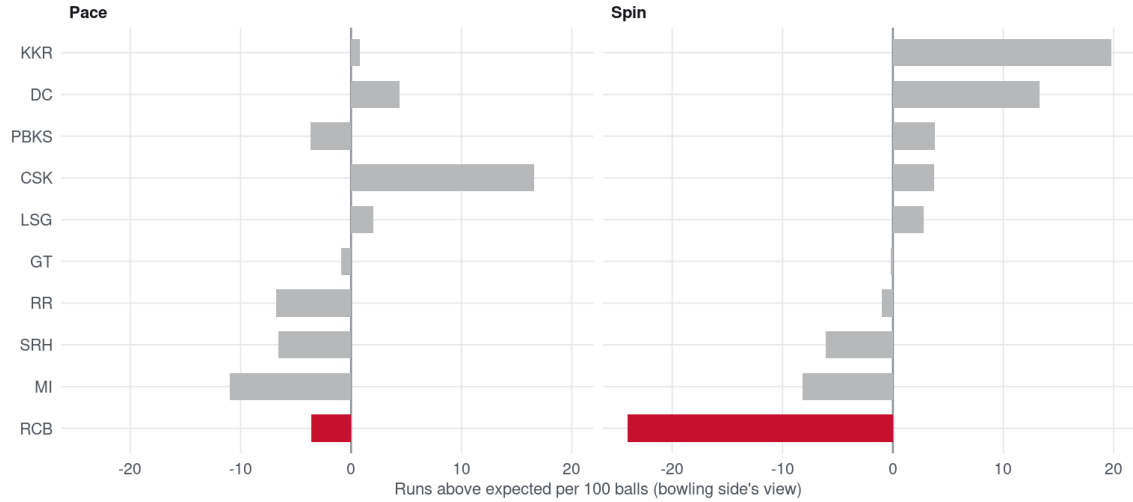
**Most of the aggregate 'can't bowl' story is the ground**

All phases, both seasons




**Middle overs, overs 7-15: spin is where it goes wrong**

Both seasons. Venue, over, wickets and batter set-ness already in the expectation.



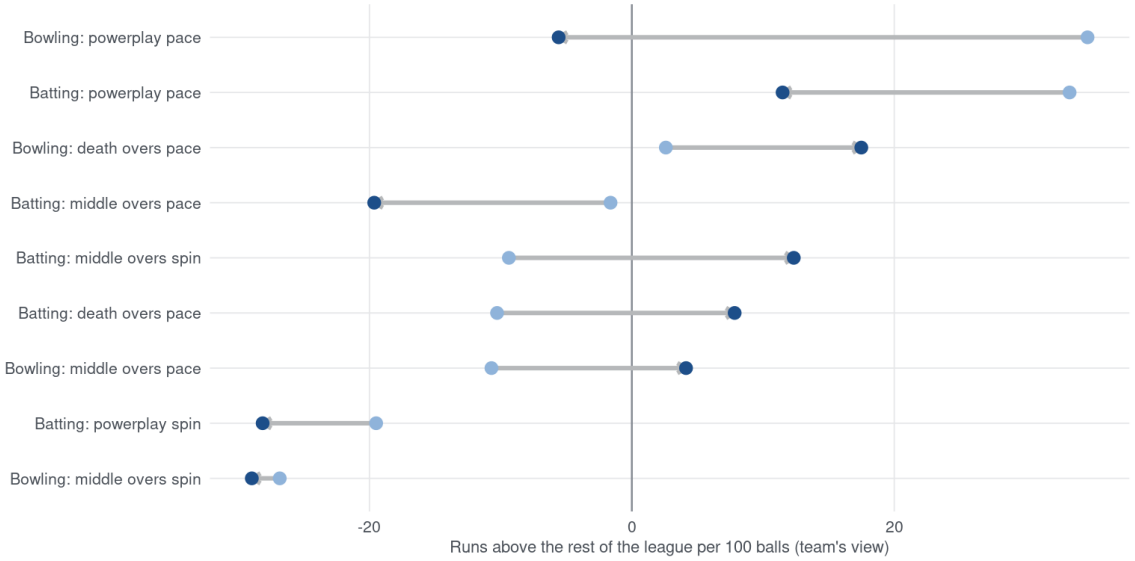

*Bowling effect +5.3 → −0.6 per 100 balls with ground fixed effects. Middle-overs spin −24.1 against expectation; every other side +3.8.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A8 · What 2023 said, and where the money went

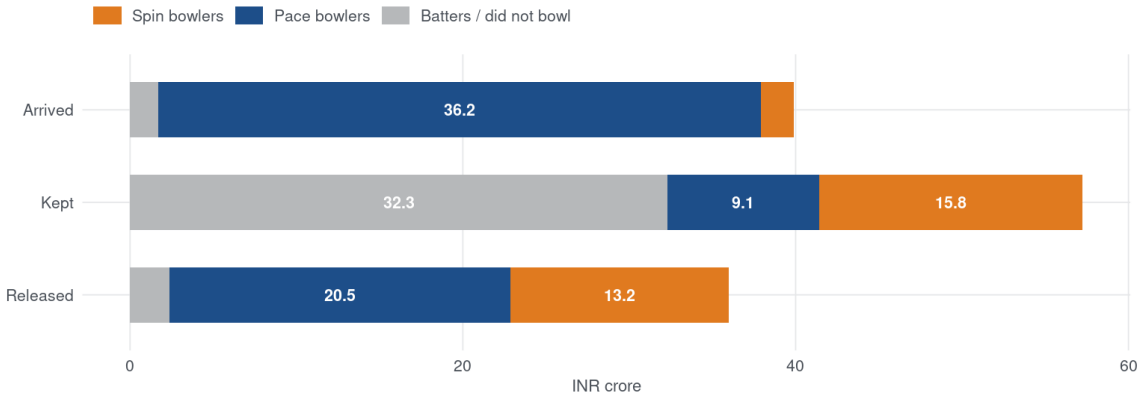
**Most of the team's weak cells moved between seasons. Middle-overs spin did not**

Runs per 100 balls against the rest of the league; light dot 2023, dark dot 2024 (arrow = direction of change)




**The weakness was spin; the new money went to pace**

Contract value (INR crore) of players released, kept and brought in for 2024, grouped by what they bowl (if they bowl)



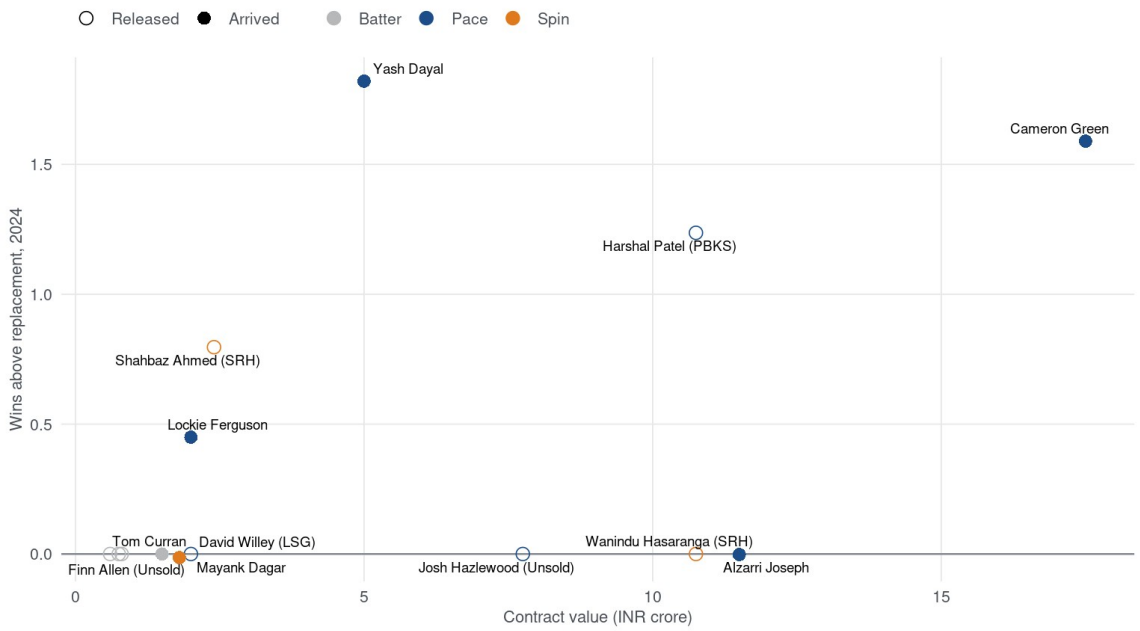

*Most weak cells moved between seasons; middle-overs spin did not (−26.8 then −29.0). New money: ₹36.2 cr on pace, ₹2.0 cr on spin.*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A9 · The decision audit: outcomes and the market

**Arrivals produced more than the players let go, at a high price per win**

2024 WAR against contract value. Released players are shown at their old price, with their 2024 team




**One good season in the middle overs is a weak guide to the next (r = -0.22)**

Regular middle-overs spinners (150+ balls in 2023, 60+ in 2024), runs saved per 100 balls. None of them was in the Dec 2023 auction



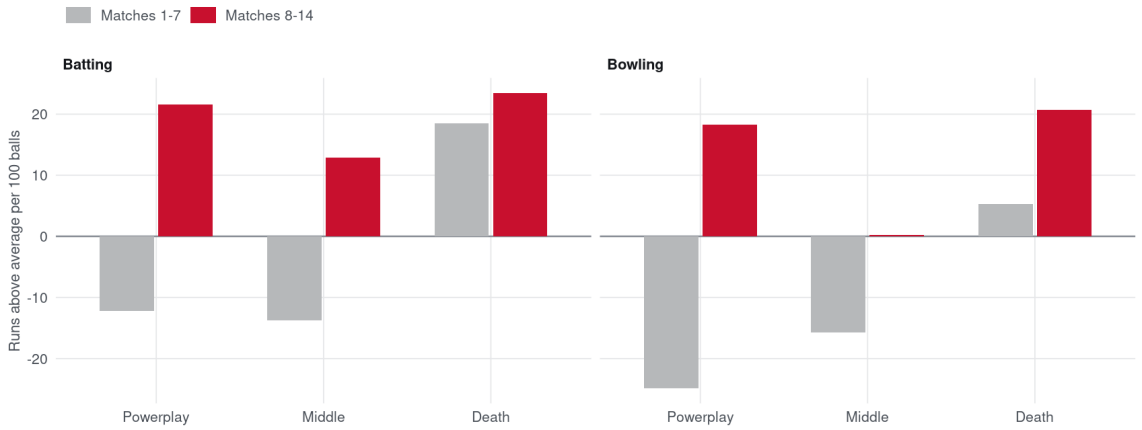

*Arrivals: 4.6 WAR at ₹8.72 cr per WAR. None of 2023's regular middle-overs spinners was in the auction; the route was a trade. Axar Patel would have been worth 1.5 wins (hindsight).*

APPENDIX — NOT COUNTED IN SLIDE LIMIT

# A10 · Inside 2024, and what the gap cost

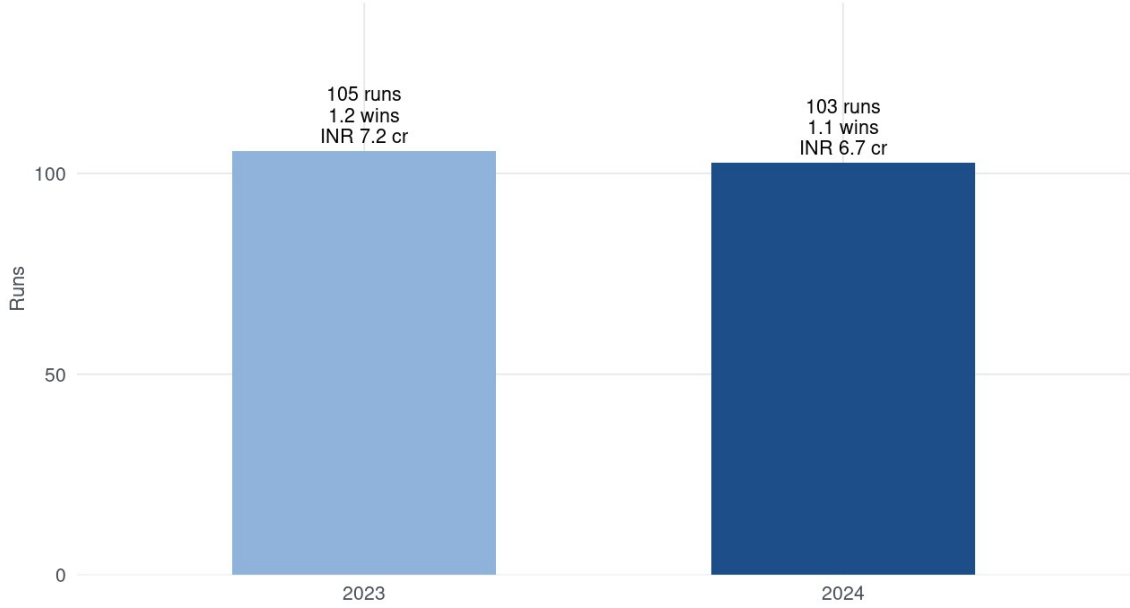
**The 2024 turnaround showed up in almost every phase, most of all in the powerplay with the ball**

Runs above average per 100 balls, team's point of view, first seven league games against the last seven



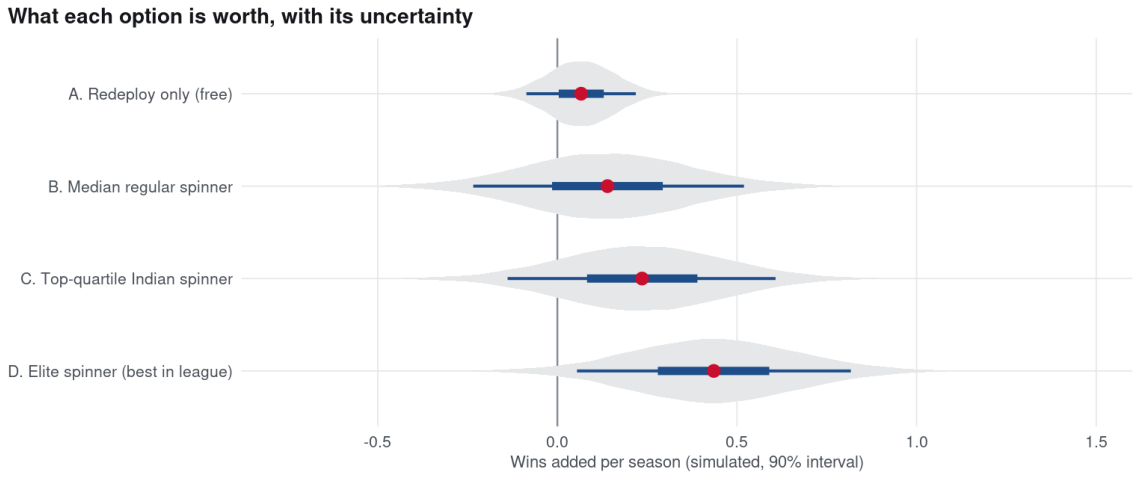

**What the middle-overs spin gap cost, each season**

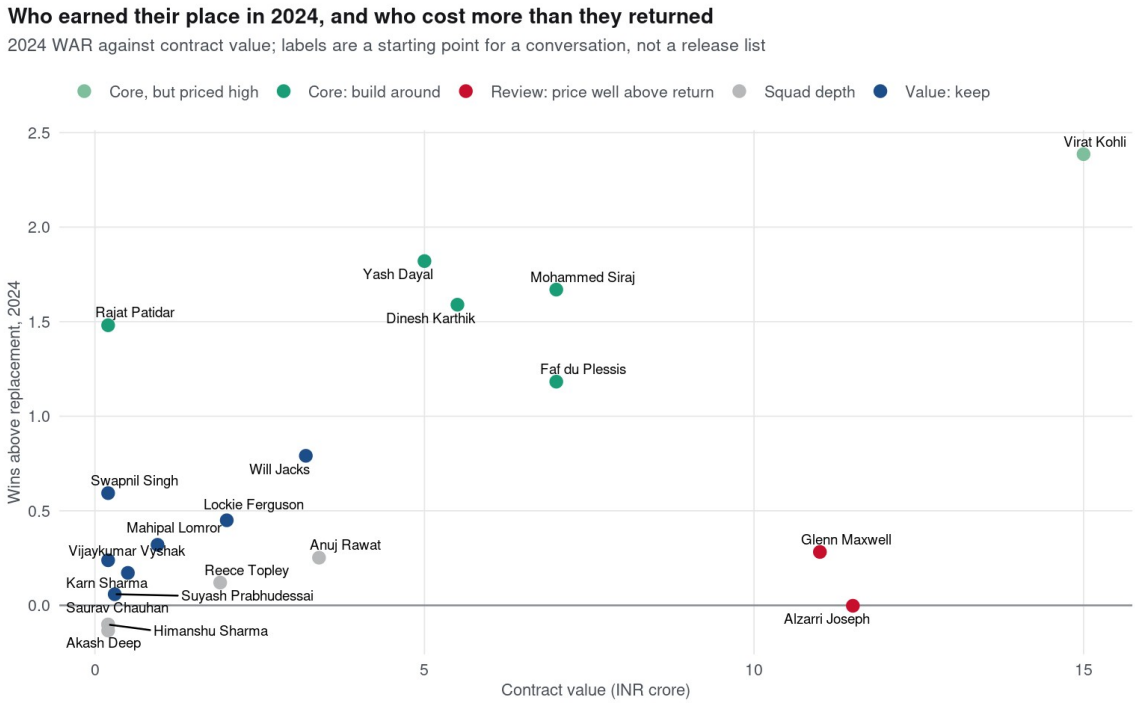
Runs conceded above the rest of the league in the team's middle-overs spin balls




*1 win in 7, then 6 in 7: the turn came in the powerplay (wickets 4 → 18). The spin gap cost 105 and 103 runs.*

# A11 · What next: options and the squad




*On shrunk projections a top-quartile Indian spinner adds 0.24 wins (85% chance positive); up to 0.8 if the gap repeated. Right: 2024 WAR against price.*